



Cambridge IGCSE™

DESIGN & TECHNOLOGY

Paper 1 Product Design

0445/12

May/June 2026

1 hour 15 minutes



You must answer on the two pre-printed A3 answer sheets.

You will need: Two A3 pre-printed answer sheets (enclosed)
Standard drawing equipment
Coloured pencils

INSTRUCTIONS

- Answer **one** question.
- Use an HB pencil for any drawings and a black or dark blue pen for any writing.
- Write your name, centre number and candidate number in the space on **both** pre-printed answer sheets.
- Answer in the space provided on the answer sheets.
- Do **not** use an erasable pen, staples, paper clips, glue, or correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.
- You may use standard drawing equipment, including coloured pencils.
- At the end of the examination, hand in your named A3 answer sheets. Do **not** fasten them together and do **not** punch holes in the sheets or tie with string.

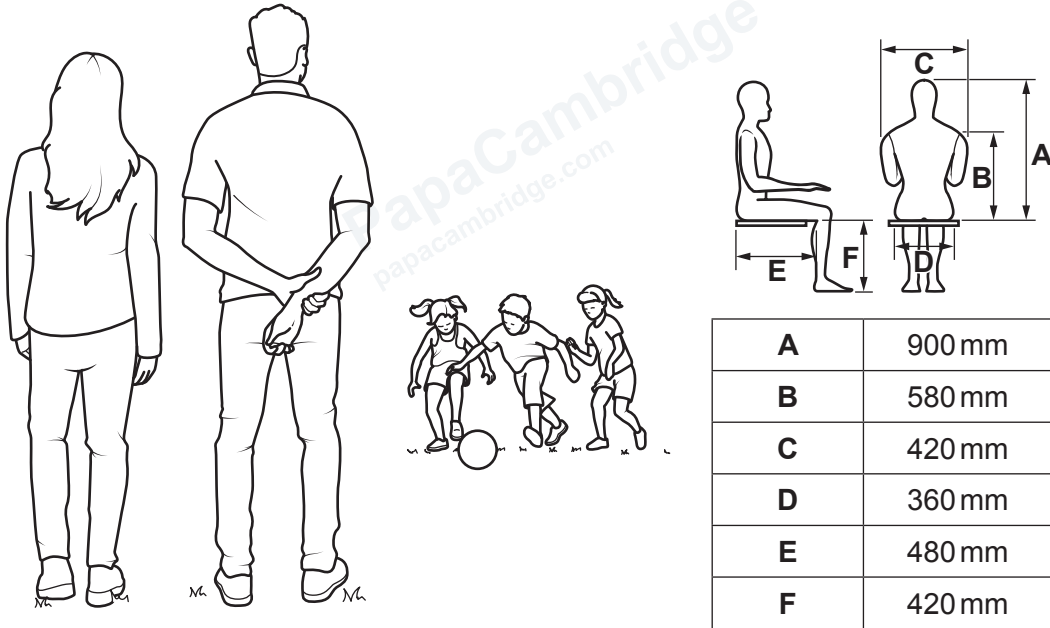
INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].
- All dimensions are in millimetres unless otherwise stated.

This document has 4 pages.

Answer **one** question only on the A3 pre-printed answer sheets provided.

- 1 Play parks are often used by families with young children.

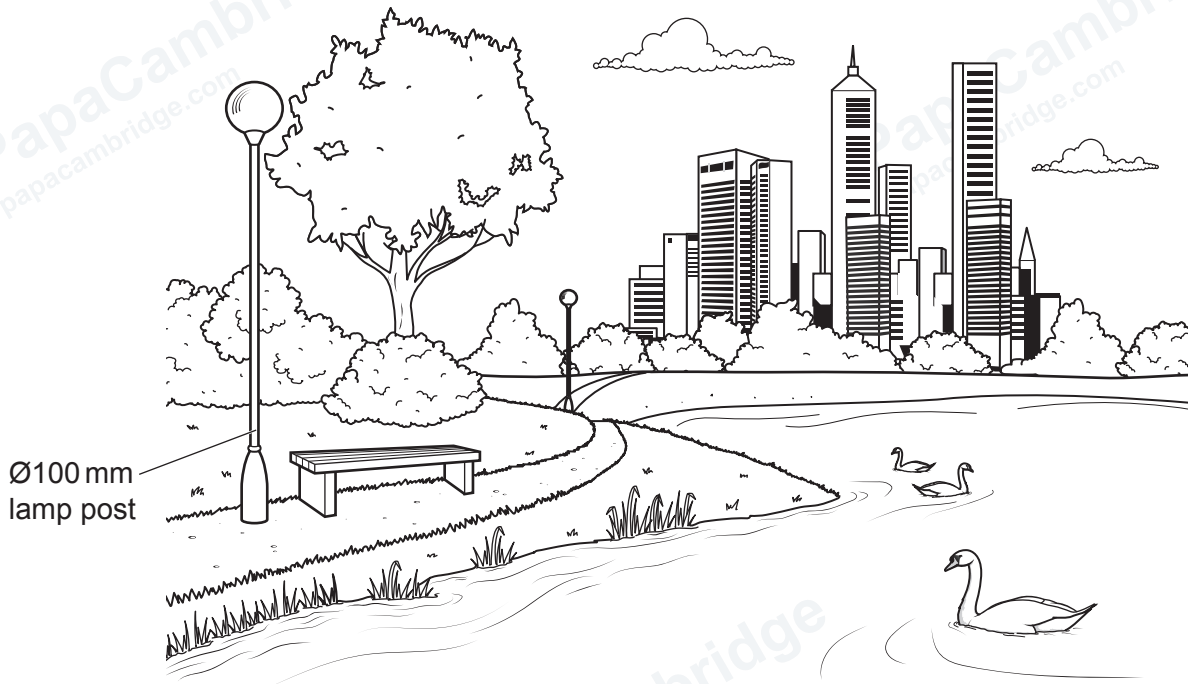


anthropometric data for
the average adult

Design a seat that can be used by a parent or carer whilst watching their children play. The seat must be lightweight and foldable so that it can easily be carried.

- List **four** additional points about the function of such a seat that you consider to be important. [4]
- Use sketches and notes to show **two** ways of making a length of resistant material lighter in weight, without altering the dimensions. [4]
- Develop and sketch **three** separate ideas for the seat. [12]
- Evaluate your three ideas. Choose **one** idea to develop further and justify your choice. [8]
- Draw, using a method of your own choice, a full solution to the design problem. Include construction details and important dimensions. [12]
- Suggest **two** suitable specific materials for the solution you have drawn in 1(e) and give reasons for your choice. [4]
- Outline a method that could be used to manufacture **one** part of your solution drawn in 1(e). Include the names of the tools used. [6]

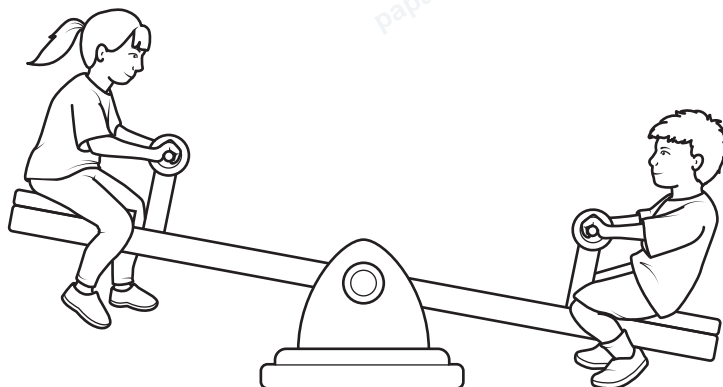
- 2 Lakes in parks can be dangerous and must **not** be used for swimming.



Design a sign that will warn people that they must **not** swim in the lake shown. The sign must fasten to the Ø100 mm lamp post and **not** use words to show the warning.

- (a) List **four** additional points about the function of such a sign that you consider to be important. [4]
- (b) Use sketches and notes to show **two** methods of warning people of a danger without the use of words. [4]
- (c) Develop and sketch **three** separate ideas for the sign. [12]
- (d) Evaluate your three ideas. Choose **one** idea to develop further and justify your choice. [8]
- (e) Draw, using a method of your own choice, a full solution to the design problem. Include construction details and important dimensions. [12]
- (f) Suggest **two** suitable specific materials for the solution you have drawn in 2(e) and give reasons for your choice. [4]
- (g) Outline a method that could be used to manufacture **one** part of your solution drawn in 2(e). Include the names of the tools used. [6]

- 3 Play equipment is provided in parks to entertain children.



Design a device that will indicate in a fun way when a seesaw is in a horizontal position. The device must temporarily attach to a seesaw.

- (a) List **four** additional points about the function of such a device that you consider to be important. [4]
- (b) Use sketches and notes to show **two** methods of detecting or sensing a change in angle of a horizontal surface. [4]
- (c) Develop and sketch **three** separate ideas for the device that attaches to a seesaw. [12]
- (d) Evaluate your three ideas. Choose **one** idea to develop further and justify your choice. [8]
- (e) Draw, using a method of your own choice, a full solution to the design problem. Include construction details and important dimensions. [12]
- (f) Suggest **two** suitable specific materials for the solution you have drawn in 3(e) and give reasons for your choice. [4]
- (g) Outline a method that could be used to manufacture **one** part of your solution drawn in 3(e). Include the names of the tools used. [6]

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